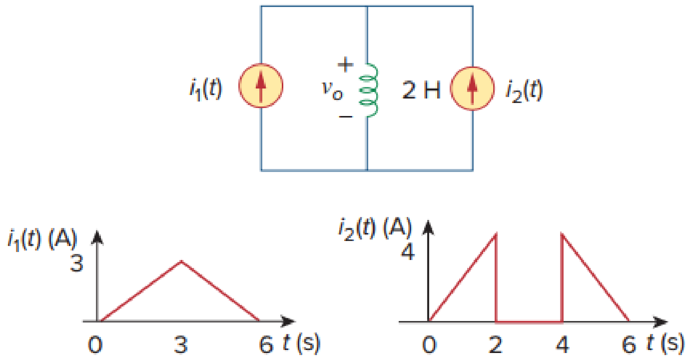
Problem

In the circuit of Fig. 6.85, sketch *vo*.

Fig. 6.85:



Step-by-step solution

Step 1 of 2

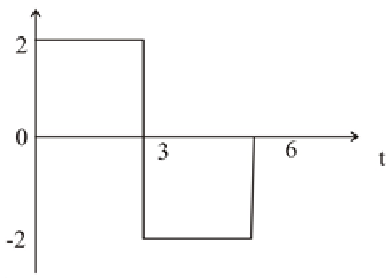
We apply superposition principle and let

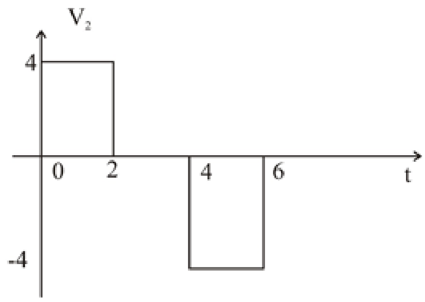
$V_0 = V_1 + V_2$

Where V_1 and V_2 are due to i_1 and i_2 respectively

$$V_1 = L\frac{di_1}{dt} = 2\frac{di_1}{dt} = \begin{cases} 2 & 0 < t < 3 \\ -2 & 3 < t < 6 \end{cases}$$

$$V_2 = L\frac{di_2}{dt} = 2\frac{di_2}{dt} = \begin{cases} 4 & 0 < t < 2 \\ 0 & 2 < t < 4 \\ -4 & 4 < t < 6 \end{cases}$$





Step 2 of 2

Adding V_1 and V_2 gives $V_{0,}$ Which is shown below.